Replication of Study "Perceptual mechanisms that characterize gender differences in decoding women's sexual intent" by Farris, Treat, Viken & McFall (2008, *Psychological Science* 19*:* 348-354)

Angela S. Attwood[1,2,3], Andy Woods[3,4], Kayleigh Easey, Ian S. Penton-Voak[3], Marcus R. Munafò[1,2,3]

[1] MRC Integrative Epidemiology Unit (IEU) at the University of Bristol, United Kingdom
[2] UK Centre for Tobacco and Alcohol Studies, United Kingdom
[3] School of Experimental Psychology, University of Bristol, United Kingdom
[4]Xperiment, Surrey, United Kingdom

Introduction

This study is part of a wider collaborative project that is investigating the reproducibility of scientific findings in psychological research – the Reproducibility Project: Psychology. A sample of studies has been selected for replication from three psychological peer-reviewed journals published in 2008. The current study is a replication of Farris, Treat, Viken and McFall (2008a).

There is ambiguity in distinguishing sexual interest from other positive (platonic) social signals, such as smiling, eye contact and increased physical proximity, particularly among males. Men reliably rate female targets as having greater sexual intent than women (Farris, Treat, Viken, & McFall, 2008b). Consistent findings have been reported in survey research, with 67% of women reporting experiencing male misperception of their sexual intent (Abbey, 1987). However, the underlying mechanisms of this effect are not known. One suggestion is that males are more sensitive to sexual intent signals. An alternative explanation is that males display a perceptual bias, whereby they require fewer confirmatory cues before labelling intent as sexual.

Farris et al. (2008a) investigated the causal basis of this effect, using an experimental paradigm that differentiated between perceptual sensitivity and bias in male and female participants. They reported that male participants were less sensitive to the distinction between friendliness and sexual interest than women, suggesting that gender differences were due to perceptual insensitivity in men. The aim of the study is to test the reproducibility of the primary finding that males show less sensitivity in distinguishing between friendliness and sexual interest than women, (F[1, 278] = 123.38, p <.001). Data analysis will involve the same statistical choice-prediction model as Farris et al (2008a), which provides estimates of sensitivity.

Methods

Power Analysis

The original study reported an effect size of η_p^2 = .326 for the comparison in sensitivity between male and female participants. Power calculations indicate that sample sizes of 19, 24 and 29 are required to achieve 80, 90 and 95% power respectively, at an alpha of .05. However, due to the degree of discrepancy between the projected sample sizes and that of the original study, we plan to recruit 144 participants.

Planned Sample

Male and female heterosexual undergraduates (n = 144) will be randomly recruited from the University of Bristol through existing email lists and poster advertisements. Interested parties

will be sent the information sheet in advance of the study session and asked to contact the research co-ordinator again if they would like further information or would like to sign up for the study. The original study did not balance numbers of males and female recruited and we will use the same random sampling approach. Participants will be reimbursed £5 at the end of the session.

Materials

Images for the identification task were obtained from the original study authors. These comprised 280 full-body images of females showing one of four emotions (friendly, sexually interested, sad, rejecting), with equal numbers of images per emotional category. For a full description of the images and the piloting procedures, see Farris et al (2008a).

The task was reprogrammed in EPrime (version 2.0 Professional, Psychology Software Tools, Inc.) by the research team, using the task parameters detailed in the original paper. Two task versions were programmed with different image presentation times (500 ms, 3,000 ms), and participants will be randomised to complete one of the two task versions. The task will begin with an instruction screen that prompts participants to start when they are ready. At the beginning of each trial, a fixation cross will appear centrally on screen for 250 ms. The image will then replace the fixation cross and will remain on screen for either 500 or 3000 ms (depending on task version). The next screen will ask participants to identify whether the woman in the previous image was “friendly”, “sexually interested”, “sad” or “rejecting” using the keyboard keys 1, 2, 3, 4 respectively. Each of the 280 images will be presented once, across four blocks of 70 randomly chosen images. There will be a 30 second break between blocks.

Procedure

Upon arrival, participants will be given a copy of the information sheet to read again if they wish, and will be asked to provide written informed consent. Participants will be seated alone in a test laboratory andwill be randomly assigned to complete the 500 ms or 3000 ms presentation time version of the task. At the end of the session they will be given a debriefing sheet and reimbursed for their participation.

Analysis Plan

The original article reported estimates of two choice parameters, sensitivity and bias. As the aim of this study was to replicate the sensitivity finding, we will compute and analyse sensitivity estimates only. The parameter (α) “provides a quantitative estimate of an observer’s ability to perceptually distinguish between categories”. The parameter will be calculated per the author’s instructions in the original paper. It can be calculated using the formula (Figure 1) where *H* is the proportion of hits and *FA* is the proportion of false alarms. A higher α score represents a better discrimination score.

$$\alpha = \sqrt{\frac{H(1 - FA)}{(1 - H)FA}}$$

Figure 1. Sensitivity formula for two alternate forced choice tasks.

For forced choice tasks with two or more alternative choices, this formula (Figure 2) is modified to calculate ln(α) as the dependent sensitivity variable. This modified formula is shown below where *m* is the number of response categories, $P(R_i|S_i)$ is the probability of a

correct response to stimulus *i*, and $P(R_j|S_j)$ is the probability of an incorrect response to stimulus *i*.

$$\ln(\alpha) = \frac{\ln\left[\prod \frac{P(R_i|S_i)}{P(R_j|S_i)}\right]}{\sqrt{2}m(m-1)}$$

Figure 2. Sensitivity formula for multiple alternate forced choice tasks.

The ln(α) values for men and women will be subjected to the same general linear model as the original paper, which includes gender, clothing style (i.e., individuals were either dressed in a provocative or conservative style) and presentation time as predictors. The model will also include an affect factor to discriminate two types of affect: positive (friendliness versus sexual interest) versus negative (sadness versus rejection). The main effect of interest is the comparison between positive effect sensitivity (i.e., the distinction between friendliness and sexual interest) in men and women.

Differences from Original Study

As our School comprises predominantly female undergraduates, we will recruit from across the University to help ensure a fairer distribution of males and females. This means we cannot offer course credits as in the original study. Instead, participants will reimbursed £5 on completion of the study..

A random sampling approach has been applied in this replication, which could yield a different participant demographic. However, like the original study recruitment will be restricted to undergraduate students. As the University of Bristol also comprises a predominantly white student cohort, we expect to achieve comparable participant demographics using this random sampling approach.

(Post Data Collection) Methods Addendum

Actual Sample

Undergraduate students (n = 144, 45% male) from the University of Bristol completed the study. Participants were aged between 18 and 26 years (*M* = 20, *SD* = 1). The majority of participants were of European descent (78%), with the remaining participants being of Asian (21%) and African American (1%) descent. All participants completed the study and no outliers were identified. Therefore the final sample size in analyses is 144.

Differences from pre-data collection methods plan

Due to the clothing time variable not being coded at the time of extraction, secondary analyses including clothing type were not conducted. However, these effects were not the focus of the replication attempt.

Results

Data preparation

The target effect of this replication was the finding that males were less sensitive to the distinction between friendliness and sexual interest than females. In order to calculate sensitivity scores, hits and miss data were extracted for each condition (i.e., sexual interest,

happiness, sadness, rejection). The sensitivity formula (Figure 2) was then applied to each set of condition scores, which yielded four sensitivity scores (one for each condition).

Confirmatory analysis

The effect identified for replication was the interaction between gender and positive sensitivity, which comprises a subset of the data. Therefore, we first ran the full factorial model (see below) as planned, in order to ascertain if there was evidence of a three-way interaction (gender × affect × sexual intent), which would support post-hoc analyses.

A 2 × 2 × 2 × 2 general linear model was conducted with affect (positive, negative) and sexual intent (present, absent) as within-subjects factors (see Table 1) and gender (male, female) and presentation time (500 ms, 3000 ms) as between-subjects factors. There was no evidence of a gender x affect x sexual intent interaction (F[1,140] = <.001, p = .98, η_p^2 <.001).

Table 1: Within-subject dependent variables within factorial ANOVA.

		Factor 2: Sexual intent	
		Present	Absent
Factor 1: Affect	Positive affect	Sexually interested	Friendly
	Negative affect	Rejecting	Sad

Despite no evidence of a three-way interaction of interest in the full factorial model, we continued with post-hoc analyses that would directly test the effect targeted for replication. We used a 2 × 2 general linear model with positive affect sensitivity (friendliness, sexually interest) as a within-subjects factor and gender as a between-subjects factor. There was no evidence of an interaction (F[1,142] = 1.19, p = .28, η_p^2 =.008), indicating that there were no differences between men and women in their distinction of friendliness and sexual interest (see Figure 1).

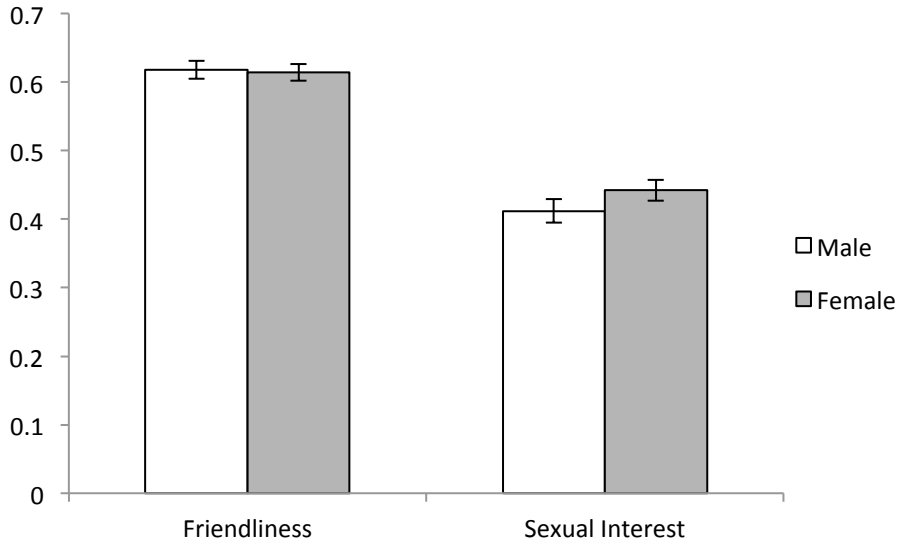


Figure 1. Mean sensitivity (+/- SE) for friendliness and sexual interest in male and female participants

Exploratory analyses

No additional exploratory analyses were conducted, although we report additional outcomes from the full factorial model. We observed significant affect × presentation time (F[1,140] = 14.70, p <.001, η_p^2 =.09) and affect × sexual intent (F[1,140] = 62.20, p <.001, η_p^2 =.31) interactions, with the mean difference between positive and negative affect being greater in the 500 ms condition (MD = 0.17 versus MD = 0.09), and the mean difference between sexual intent sensitivity being greater in the positive affect versus negative affect condition (MD = 0.19 versus MD = 0.04). There were main effects of affect (F[1,140] = 177.10, p <.001, η_p^2 =.56), sexual intent (F[1,140] = 120.54, p <.001, η_p^2 =.46) and presentation time (F[1,140] = 25.38, p <.001, η_p^2 =.15), indicating greater sensitivity for positive (M = .52, SD = .08) versus negative (M = 39, SD = .10) affective images, non-sexual (M = .51, SD = .08) versus sexual (M = .40, SD = .08) intent images and following longer (M = .48, SD = .06) versus shorter (M = .43, SD = .06) presentation times. There was no evidence of any other main effects or interactions (ps >.11).

Discussion

Summary of Replication Attempt

This study aimed to replicate the finding that males show reduced sensitivity in distinguishing between friendliness and sexual intent. We failed to replicate the finding previously reported by Farris et al (2008a). An explanation for this lack of replication may be due to cultural differences of participants based in the USA versus the UK. Alternatively, the different time periods during which the data were collected may have driven the differential findings, due to the images appearing dated in the current study. Age and ethnicity were well matched so are not expected to have influenced the failure to replicate the target finding.

Commentary

The pattern of means were in a direction consistent with the original finding, however based on effect sizes observed in the original study, this study was adequately powered to replicate a similar effect.

References

Abbey, A. (1987). Misperceptions of friendly behavior as sexual interest: a survey of naturally occurring incidents. *Psychology of Women Quarterly, 11*, 173-194.

Farris, C., Treat, T. A., Viken, R. J., & McFall, R. M. (2008a). Perceptual mechanisms that characterize gender differences in decoding women's sexual intent. *Psychol Sci, 19*(4), 348-354. doi: 10.1111/j.1467-9280.2008.02092.x

Farris, C., Treat, T. A., Viken, R. J., & McFall, R. M. (2008b). Sexual coercion and the misperception of sexual intent. *Clin Psychol Rev, 28*(1), 48-66. doi: 10.1016/j.cpr.2007.03.002